OLYMPUS ENERGY PVT LTD

The Registrar,
National Electric Power Regulatory Authority (NEPRA)
OPF Building G-5/2,
Islamabad.

October 25, 2011

Subject: **Olympus Energy (Pvt.) Limited-20 MW Lucky Hydropower Project ("Project") at Marala Headworks**

Respected Sir,

Reference Tariff Petition ("Petition") submitted for the determination of feasibility stage Tariff for the Project. We would like to present following clarifications regarding our submission to your good office.

1. The petition for the determination of tariff was initially filed in August 2010 to the Authority. The Authority while hearing the Petition made certain observations regarding the difference in the cost estimates presented and approved in the feasibility study and those presented in the said Petition. The Authority further advised the Sponsors to get the revised cost estimates approved by the Panel of Experts (POE) appointed by PPDB in the Revised Feasibility Study.

2. As directed by the Authority, the Sponsors prepared the Revised Feasibility Study and the contents of the same were thoroughly deliberated and analyzed by POE. There were many revisions until the Revised Feasibility Study was approved by the POE.

3. From the abovementioned facts, it is evident that the current Petition is the continuation of the original Petition submitted in August 2010.

4. We would like to present our humble request before your good office that considering the facts mentioned above, the Petition may please be considered as continuation of previous submission for the purposes of prescribed fee and the requirement of new fee submission may please be waived off.

Sir, as you may kindly appreciate, Sponsors have put in a lot of time and effort to develop this project based on indigenous resources and would therefore seek patronage of the Authority through your goo office.

For n.a.w.
AD (MR)
PA
27/10/11

Best regards

(Malik Fida Hussain)
Manager Project

House # 50-B, Block D-1, Gulberg III, Lahore, Pakistan. Tel: (042) 5871784/5762479 Fax: (042) 5712008

ORIGINAL

OLYMPUS ENERGY PVT LTD

The Registrar
National Electric Power Regulatory Authority (NEPRA)
OPF Building,
Shahrah-e- Jamhuriat, G-5/2,
Islamabad.

October 17, 2011

For information & n/a pl
AD(MR) + PA

Subject: **Tariff Petition for 20Mw Lucky Hydropower Project (Project) at Marala Headworks, Sialkot, Punjab, Pakistan**

Dear Sir,

1. The letter of Interest (LOI) to conduct the feasibility study of the Project was issued to Olympus Energy (Pvt.) Limited ,("Petitioner") by the Punjab Power Development Board (PPDB) under Punjab Power Generation Policy ,2006.The Policy is mainly the adoption of Power Policy 2002 announced by the Federal Government.

2. Sponsors have conducted the feasibility study through reputable Consultants and the same has been approved by the Panel of Experts. Upon the directives of the Authority, feasibility study has been amended to include certain cost items and the feasibility study after the incorporation of the same has been approved by POE (Letter enclosed).

3. The Authority has issued Generation License No .IGSPL /24/2010 on 26th January 2010.

4. This Tariff Petition is being submitted for the determination of Generation Tariff as per the Guidelines for the Determination of Tariff announced by GOP and the " Mechanism for Determination of Tariff for Hydropower Projects" announced by NEPRA on 18th July,2008.

Original Received alongwith two copies on 25-10-11

House # 50-B, Block D-1, Gulberg III, Lahore, Pakistan. Tel. (042) 5871784/5762479 Fax: (042) 5712008

5. It is respectfully submitted to the Authority to determine Generation Tariff for the Electricity produced by the Project in accordance with para 4 and with due consideration to the assumption forming the basis of the Petition.

Thanks and best regards

(Laeeq Ahmad)
Chief Executive Officer

OLYMPUS ENERGY PVT LTD

RESOLUTION OF THE BOARD OF DIRECTORS OF THE COMPANY HELD ON August 15, 2011

Resolved, that the company file its tariff petition with National Electric Power Regulatory Authority (NEPRA)

FURTHER RESOLVED that Mr. Laeeq Ahmed Sheikh the Chief Executive of the company is hereby authorized for and on behalf of the Company to sign all the necessary documentation, pay the necessary fees, appear before NEPRA as needed and to do all such acts necessary for processing and completion of the Petition.

Signed

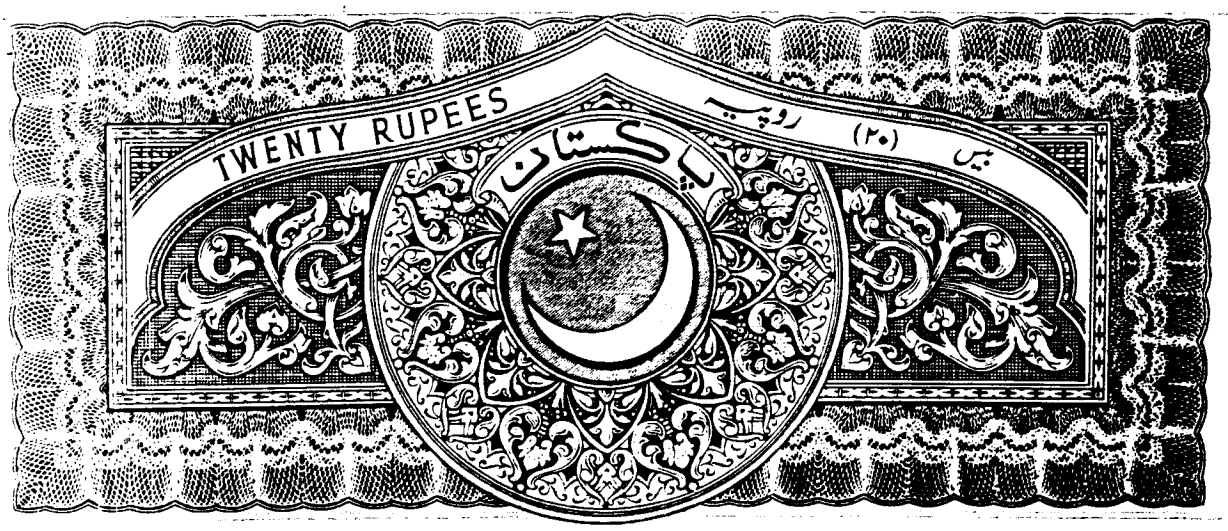
Laeeq Ahmed Sheikh
Chief Executive

Signed

Sikander Laeeq
Director

House # 50-B, Block D-1, Gulberg III, Lahore, Pakistan. Tel: (042) 5871784/5762479 Fax: (042) 5712008

PAKISTAN 20 RS.



BEFOPRE THE NATIONAL ELECTRIC POWER REGULATRY AUTHORITY

AFFIDAVIT

I, Laeeq Ahmed Sheikh, Chief Executive, Olympus Energy (Pvt.) Ltd being duly authorized representative/attorney of Olympus Energy (Pvt.) Ltd, hereby solemnly affirm and declare that the contents of the accompanying Tariff Petition dated 15.08.2011 including all supporting documents are true and correct to the best of my knowledge and belief and that nothing has been concealed.

I also affirm that all further documentation and information to be provided by me in connection with the accompanying petition shall be true to the best of my knowledge and behalf.

DEPONENT

No. PPDB/ 1334 /2011

PUNJAB POWER DEVELOPMENT BOARD
ENERGY DEPARTMENT
1st Floor, Central Design Building,
Irrigation Secretariat, Old Anarkali, Lahore
(Ph: 042-99212794 Fax: 042-99212796)

Date 18/07/2011

M/s Olympus Energy (Pvt) Ltd
House # 50-B, Block-D-1, Gulberg-III
Lahore.

Sub: UPDATED FEASIBILITY STUDY REPORT OF 20 MW LUCKY HYDRO POWER PROJECT AT MARALA HEADWORKS

Reference: 1. Minutes of Meeting of Panel of Experts (POE) dated 23.05.2011
2. Your letter No. OEL.IR/09/171 dated 22.06.2011

As already conveyed vide our letter No.PPDB/985/2011 dated 11.05.2011 that Panel of Expert (POE) appointed by PPDB Board, has approved the updated Feasibility Study Report in respect of your project in light of direction of NEPRA vide letter No. NEPRA/TF-100/2074-76 dated August 27, 2010. Therefore, now, you may proceed further in light of Hydro Power Mechanism – July 2008 issued by NEPRA with the respect to Hydropower Project.

MANAGING DIRECTOR
Punjab Power Development Board

CC:
1. The Chairman PPDB, 10-A, Ali Block, New Garden Town, Lahore.
2. The Secretary, Govt. of Punjab, Energy Department, Lahore.

BEFORE

THE NATIONAL ELECTRIC POWER REGULATORY AUTHORITY (NEPRA)

PETITION FOR THE DETERMINATION OF FEASIBILITY STAGE GENERATION TARIFF

FOR A HYDROPOWER POWER PROJECT (Lucky Hydropower Project) OF APPROX. 20MW - GROSS

AT

Marala HEADWORKS , PUNJAB PAKISTAN

OLYMPUS ENERGY (PVT) LIMITED

ON BEHALF OF
LUCKY HYDROPOWER COMPANY (PROPOSED)
50-BLOCK D/1, GULBERG 3 LAHORE

GLOSSARY

Abbreviation	Name / Term
AY	Agreement Year
CAPEX	Capital Expenditure
CED	Central Excise Duty
COD	Commercial Operations Date
CPI	Consumer Price Index
CPP	Capacity Purchase Price
CY	Calendar Year
DC	Dependable Capacity
DCS	Distributed Control System
ECC	Economic Coordination Committee
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EPA	Environmental Protection Agency
EPC	Engineering, Procurement & Construction
EPP	Energy Purchase Price
FBR	Federal Board of Revenue
FM	Force Majeure
GOPb	Government of Punjab
GOP	Government of Pakistan
GST	General Sales Tax
IA	Implementation Agreement
IDC	Interest during Construction
IEE	Initial Environmental Examination
IPPs	Independent Power Producers
KIBOR	Karachi Inter-bank Offered Rate
LC or LoC	Letter of Credit
LDs	Liquidated Damages
LIBOR	London Inter-bank Offered Rate
MO	Maintenance Outage
MoF	Ministry of Finance
MWh	Mega Watt Hours
NDT	Non Destructive Testing
NEO	Net Electrical Output
NEPRA	National Electricity and Power Regulatory Authority
NPCC	National Power Control Center
NTDC	National Transmission & Despatch Company Ltd.
O&M	Operation & Maintenance
OEM	Original Equipment Manufacturer
PEPCO	Pakistan Electric Power Company (Private) Ltd.
PPA	Power Purchase Agreement
PPDB	Punjab Power Development Board
PPIB	Private Power and Infrastructure Board
PRI	Political Risk Insurance
ROE	Return on Equity
ROW	Right of Way
SBP	State Bank Of Pakistan
SECP	Securities and Exchange Commission of Pakistan
SO	Scheduled Outage
SOP	Standard Operating Procedures
LPC	Late Payment Charges
LTSA	Long Term Services Agreement
MI	Major Inspection
MIGA	Multilateral Investment Guarantee Agency

1. PROJECT BACKGROUND ,HISTORY and Basis for the Petition

1.1 Marala Barrage site was identified in 1980 through ranking studies carried out by WAPDA, on the Irrigation network of the country. The PPDB thus considered a Low Head Hydropower Project at Marala Barrage and issued a Letter of Interest (LOI) in favor of M/S Olympus Energy (Pvt.) Ltd. in December, 2007 vide No. MD-PPDB/H-5/489 dated 3-12-2007 to carry out feasibility study by the name of Lucky Hydropower Project at Marala Barrage. The conduct of feasibility study was closely monitored by the Panel of Experts and the feasibility study of the project was approved by the Panel of Experts (POE) appointed by PPDB.

1.2 The Sponsors submitted before NEPRA ("Authority) a petition for the determination of Tariff based on Feasibility Study under the Mechanism for the Determination of Tariff for Hydropower Projects - 2008 issued by the Authority in August 2010. However on account of difference in the cost estimates approved in the feasibility study and the cost estimates presented in the petition, the Authority ordered that Sponsors should get the cost estimates revalidated by POE by incorporating the revised cost/ additional cost items in a revised feasibility study.

1.3 The Sponsors prepared the revised feasibility as per the directives of the Authority and the revised feasibility study has been approved by the POE appointed by PPDB. Therefore the Sponsors would like to plead before the Authority for the submission of Tariff Petition based on revised feasibility study.

1.4 Under the Regulation of Generation, Transmission and distribution of Electric Power Act (Act No. XL) of 1997 (the **"NEPRA Act"**), the Authority is mandated to determine tariffs and other terms and conditions for the supply of electricity through generation, transmission and distribution.

1.5 This Tariff Petition is being filed before National Electric Power Regulatory Authority (the **"NEPRA"** or the **"Authority"**) pursuant to Rule 3 of the NEPRA (Tariff Standards and Procedure) Rules, 1998, read with paragraph 1.3 of the Guidelines for Determination of Tariff for Independent Power Producers issued by the Government of Pakistan in November 2005 and the applicable provisions of the Government of Pakistan's Policy for Power Generation Project, 2002 (the **"2002 Power Policy"**) and any amendments as agreed with the Petitioner during detailed negotiations on the Power Purchase , Water use license and Implementation Agreements. The cost estimates presented in the Tariff Petition are based on the fact that the provisions of Mechanism for the Determination of Tariff for Hydropower Projects-2008 shall be applicable

2. INTRODUCTION TO THE PROJECT

2.1 Name and Address of the Petitioner

Registered Office:
Olympus Energy (Pvt) Limited for and on behalf of
Lucky Hydropower (Pvt) Limited –Proposed
50-B, Block D/1, Gulberg 3, Lahore

Business Address:
As above

2.2 Authorized Representatives of the Petitioner

Mr. Laeeq Ahmad, Chief Executive Officer, Olympus Energy (Pvt) Limited,

Chief Executive, LUCKY Hydropower Company (Proposed)

2.3 Ownership of the Company and Brief Profiles of the Shareholders

Lucky Hydropower Project is being sponsored by the current shareholders of Olympus Energy (Pvt) Limited. Shareholder of Lucky Hydropower Company (proposed) is currently Olympus Energy (Pvt) Limited .Other shareholders may be added prior to financial close. On completion, Lucky Hydropower Project will be one of the largest power projects of on irrigation networks in Pakistan.

Brief profile of Olympus Energy (Sponsors of Lucky Hydropower Project) is provided below:

2.3.1 *Olympus Energy (Pvt.) Limited*

Olympus Energy (Pvt.) Ltd (OEPL) is a Private Limited Company, incorporated in Pakistan on 15th March 2006 under the Companies Ordinance, 1984. Its registered office is located at 69-B-1, Gulberg III, Lahore. OEPL was formed to develop, design, construct own, posses, manage, and maintain electric power generation plant(s) (hydel/thermal/renewable) in Pakistan, and in connection therewith to engage in the business of generation, transmission, sale, supply and distribution of electricity with Pakistan and to do all and everything necessary, suitable, proper, incidental or conducive to the accomplishment of this object and to do every other act or thing incidental or appurtenant to or arising out of or connected with this object.

Mr. Laeeq Ahmed Sheikh, Chief Executive Officer OEPL, is very well knows personality in Pakistan. Mr. Sheikh has completed its Master degree from the Wharton School, University of Pennsylvania, USA. His studies emphasis on international business and marketing including strategic planning, foreign markets and risk management with special emphasis on counter trade development in LDCs. Has Served as Member with Privatization Commission, Government of Pakistan from 1994-96, Director of the Board of Oil and Gas Development Company Limited (formerly Oil & Development Corporation). He is currently on the board of Privatization Commission, Government of Pakistan.

2.4 Financial Structure

The planned financing of the Project is through long-term debt (80%) and equity (20%). Lucky Hydropower Company (proposed) Shareholders will provide the equity for funding the Project. The Project will utilize conventional and Islamic long-term debt raised through an optimal mix of available financing options. The currently contemplated financing structure is:

1	Capital Cost	USD 41.4661 million
2	Debt Equity ratio	80/20
3	Debt	USD 33.1729million
4	Equity	USD 8.2932 million

It is assumed that all the lending for the Project shall be arranged locally and the tariff is calculated on the basis of KIBOR plus a margin/spread of 4.5%. In case of any change In the debt composition, the debt servicing component and the resultant Tariff shall be adjusted accordingly, as has been mentioned in the Determinations made by Authority in case of various other projects.

In case, the sponsors are able to arrange lending for the project below the benchmark rate of KIBOR plus 4.5%, the amount of the savings shall be shared between the sponsors and the Power purchaser as per the GOP guidelines for the determination of Tariff in the following ratio:

Power Purchaser: Sponsor

60 : 40

The financing structure and the associated terms in this proposal are based on responses received from the financial markets. In today's volatile market environment, it is critical for the Authority (NEPRA) to recognize that the financing structure and associated terms are fully subject to adverse changes in the financial markets. This financing structure assumes that full indexation for foreign exchange (for foreign loans, if any) and interest rate changes related to the debt financing will be allowed under the tariff.

Development of Low HEAD Hydel Projects

The Energy Crises in the country demands an immediate relief by adding power supplies in the national grid system. The power supplies if come through the hydropower development will be very promising for its environmental friendly status as well as its cheaper rates.

The Low head hydropower development when compared to the high head projects are less economical but have the following advantages.

- The time for planning and implementation is shorter and thus can be pushed to meet immediate needs.
- Project locations are nearer to the load centers.
- Infrastructure in the form of existing civil works and national grid in the near vicinity reduces the main cost.

A catalyst in the economic development of remote areas in the form of employment offered to the locals, utilization of the local materials available for construction and other social & economical aspects in the project area and around.

2.5 Benefits Associated With the Project

Most specifically, the following benefits associated with Lucky Hydropower Project will multiply once the Lucky Hydropower Project connects to the grid. The Project will provide much needed electricity both for the province of Punjab as well as strengthening the GEPCO grid with reliable power without any fuel cost. Other benefits are discussed below:-

2.5.1 Economical Tariff

Lucky Hydropower Project will save approximately **US$ 12.7 million** annually based on the current fuel prices and the exchange rate in terms of total billings to Power Purchaser based on actual generation when compared to an existing comparable size generation Plant utilizing imported Fuel Oil as well as the additional logistical issues with transferring oil upcountry in Pakistan. With prevailing oil prices and proposed expansion, this benefit will substantially increase.

2.5.2 Savings in Foreign Exchange

Lucky's utilization of the hydel source within Pakistan not only benefits Power Purchaser but also saves precious foreign exchange to the national economy in terms of lower fuel imports required for similar generation in oil-fired plants.

Lucky Hydropower Project will be an important and valuable asset for Pakistan. Even without a significant power shortage, the Plant would be expected to be despatched at base load in economic preference to virtually all of the oil-fired thermal power stations in Pakistan. The Savings to the national exchequer during the term of PPA (30 years) based on current fuel price and exchange rates is USD 381 million

2.5.3 Payments to Punjab Government

Lucky Hydropower Project will enhance GOPb's revenue in the form of Water Use Charge which at the current rate will be Pak Rs. 15.3 million and for the term of the PPA, the amount (without adjustment for inflation etc.) is Pak Rs. 459 million. In addition various governmental duties and levies shall be paid by the

Project (recoverable under Pass through mechanism) revenue thereby contributing to its financial strength.

2.5.4 Environmental Friendly Operations

Lucky Power Project will be environment friendly. The Environmental Impact Analysis confirms that the Project will cause no significant environmental impacts and no land or population relocations. The utilization of high efficiency low emission plant technology and the use of the existing plant site are major contributors toward achieving minimal environmental impact with low air emissions, and zero liquids discharges.

2.6 Synergies Associated with use of Existing Facilities

An addition of incremental capacity at the existing site provides several synergies specifically considering readily available land, basic infrastructure during construction; secured construction site; etc. The immediate availability of land alone is expected to save more than six months in the overall Project Development Schedule, a benefit factored into the Project development timeline.

The number of interfaces and synergies are as follows ; (1) land and access rights; (2) use of existing facilities;

2.6.1 Use of Land and Access Rights

Lucky Hydropower Project will lease/obtain the main plant site from Irrigation department and be granted full rights of access and usage.

The Project will save upfront costs for land acquisition by developing the project on leased land currently owned by irrigation department, GoPb.

2.6.2 Permanent Roads, Extra Construction and Fences

As discussed above, Lucky Hydropower Project will use the existing road to the main highway, , and certain other roads, accordingly, by not constructing this infrastructure has resulted in these costs not being assumed in the overall EPC costs for the project other than the internal and connecting roads required for the expansion.

2.6.3 Long Term Services Agreement/ Spare Parts

Lucky will sign a Long Term Services Agreement with the OEM to support the maintenance of the expansion. Costs and assumptions on the term and scope of this Agreement has been included in the overall tariff for the project.

3. TARIFF PETITION

This Tariff Petition is being submitted by Olympus Energy (Private) Limited (hereinafter the "Petitioner") to NEPRA for the 20 MW Gross low head hydropower project. The details of the project and costs are provided below.

3.1 Plant & Equipment Procurement /Civil Works

The Petitioner has based the Tariff calculations on the basis of the cost of Electromechanical and Civil works as mentioned in the feasibility study duly approved. Since at the feasibility stage it is very difficult to determine the exact EPC cost for hydropower projects, therefore NEPRA upon the advice from ECC; issued Mechanism for the Determination of Tariff for Hydropower Projects. Pertinent details are as follows:

Quoted from GoPb' Power Generation Policy

Quote:

"COST VARIATIONS (ESCALATION, RESETTLEMENT AND GEOLOGICAL RISK)

ECC of the Cabinet, through its decision dated 23rd May 2007 had interalia, decided that:

> ***"NEPRA should stop the practice of accepting EPC costs on the basis of quotations etc. Instead, they should base their determination on firm (non-reopen able) competitive price duly initialed/ signed by the IPP/EPC contractors."***

The Sponsors of Hydel Projects conveyed that it is difficult for them to obtain a firm and final cost for hydropower projects at the feasibility stage due to their site specific nature, geological risk, long construction period and environmental sensitivities, therefore, above mentioned ECC decision is not workable. In order to expedite the implementation of private sector hydropower projects, the ECC through its decision dated 22nd January 2008 decided as under:

> ***"The above decision regarding firm cost may not be applicable to hydropower Projects"*** **and that** ***"Mechanism for determination of tariff for Hydropower Projects shall be issued by NEPRA."***

NEPRA on 18th July 2008 has issued a "Mechanism for Determination of Tariff for Hydropower Projects" allowing the cost variations (Re-Openers) to be adjusted at EPC stage and/or at the Commercial Operation Date (COD). The Re-Openers allowed are as under:

Cost Variation due to geological conditions, limited to tunnel area; Civil works cost escalation, and ; Resettlement costs."

Unquote

Although experience and market exposure has been utilized to estimate the EPC Cost yet it is

assumed that the Cost variations on account of the variations mentioned above shall be accounted for at the EPC and at COD and the Tariff shall be adjusted accordingly.

3.2 The objectives

The objectives being pursued during the preparation of specifications for EPC are as follows:

- ✓ High plant efficiency and low operating cost;
- ✓ Utilization of proven design directed toward extended plant life and total system
- ✓ Reliability;
- ✓ Redundancy of specific system and control to assure a high degree of plant
- ✓ Availability;
- ✓ Operational flexibility under various operating conditions; and
- ✓ Equipment layout for easy maintenance.

4. PROJECT COST

4.1 Cost Break-Up

Following is the break-up of the project costs:-

EPC Contract /Construction Cost	**31,0430**
Development Cost	2,275
Insurance during Construction	419
Cost of loan arrangement	964
Interest During Construction (IDC)	5,651
Duties and Charges	1,113
Total Project Cost	**41,4661**

4.2 EPCC Cost

The estimates given in the feasibility study provide an EPC cost of **US$ 31.0430 Million**. The break up is provided in Annexure B (Cost Estimates).Salient features of the Project/Complex including various components of civil works as well as electromechanical equipment are mentioned in Annexure A (Project Brief). The figure includes warranty spares provided under the EPC.

4.3.1 Development Cost

The cost comprises of the costs associated with prequalification, feasibility study, field investigations, employer's office expenses before financial close, Owner's Engineer , technical and financial advisors and sponsor's office expenses during construction phase. The details are provided in Annexure B (Cost Estimates).The estimated figure is **USD 2.275 million**.

4.3.2 Insurance during Construction

Insurance during construction has been assumed @ 1.35% of the EPC Price. Any variation will be adjusted and the sponsors will provide documentary evidence in case of increase in the premium. The estimated figure is **USD 0.419 million**

4.3.3 Cost of loan arrangement

This cost comprises of the cost of due diligence by lenders, registration of documents and fees for the advisors appointed by lenders to the Project. The estimated figure is **USD 0.964 million**. The details are given in Annexure B (Cost estimates). It is pertinent to mention here that the figure is 2.9% of the loan amount. It is requested that a cap of 3% of the loan amount may be approved as has been allowed to other projects and the percentage amount should be linked with the loan amount to account for any variation in loan amount because of variation in EPC price from that conceived herein.

4.3.4 Interest during Construction (IDC)

The estimated figure is USD 5.651 million. The IDC is calculated based on a Construction period of 32 months. The interest rate used is KIBOR (13%) plus a Spread of 4.5%. The IDC figure shall be adjusted as per actual subject to provision of documentary evidence by the sponsors.

4.3.5 Duties and Other Charges

The estimated figure is USD 1,113 million. The number is arrived by the application of duties @ 5% on foreign component of EPC for duties and 3% for port clearance, handling and inland transportation.

4.3.6 Cost of Letter of Credit.

The Company is obligated to provide an LC of USD 27.5 /kW of the Contract Capacity. In addition lenders' would need an LC for the Cost overruns upto 25% of the Capital Cost of the Project. Likewise the Company shall provide LC's to Contractors LC's for EPC and O&M Contractor. The estimated amount is **USD 200,000. However as advised by the POE,its cost has not been included in the cost estimates with the assumption that actual cost incurred on the provision of LCs would be allowed subject to the provision of documentary evidence.**

4.4 Tariff Structure & Summary

The Tariff is a two-part tariff, comprising Capacity and Energy charges, based on a 30 years term. This tariff will be incorporated into the PPA to be entered into in due course between the Petitioner and CPPA. The Project is being developed on BOOT basis and at the end of the Term of PPA, it will be handed over to the Government at a nominal price of Re 1 .

This Tariff Petition contains detailed assumptions and adjustment formulae which will form part of Schedule 1 to the PPA.

The capital structure of the Project is as follows:

	US$ (000)
Total Project Cost	41,4661
Project Debt	33,1729
Project Equity	8,2932
Debt: Equity ratio	80 : 20

These numbers are based on the detailed assumptions stated in the appended assumptions and stated elsewhere in this Petition. Change in the assumptions shall result in a change in these numbers. The proposed tariff figures appended herein below are the result of a detailed financial analysis and consideration of technical, commercial and legal aspects.

4.5 Energy Charges

The Energy Charge is based on the actual net electrical output measured in kWh, and consists of:-

(i) Water Use Charges @ Re 0.15/kWh
(ii) Variable O&M component @ Re.0.193/kWh

4.6 Variable O&M

This component primarily includes lubricant consumption, , consumables, imported spare parts to be changed on normal scheduled maintenance and unscheduled maintenance. Also, it includes specialized technical services from manufacturer during maintenance of the Power Plant. The equipment has manufacturer-recommended overhauling schedules that are based on actual running hours. The actual timing of the major overhaul depends on the dispatch provided to the Power Plant. The gas turbines are expected to be made in France; therefore, the spare parts and specialized technical services will be supplied from Europe. Based on the aforesaid, the variable O&M (foreign component) will be indexed to the European CPI. This tariff component will also be adjusted for currency indexation on a quarterly basis.

4.7.1 Non -Debt Capacity Component

This component represents the Fixed O&M costs (Foreign and Local), insurance, financing cost of working capital, return on equity, return on equity during construction and withholding tax. A summary of the Capacity charges is provided below:

4.7.1.1 Fixed O&M Local and Foreign

The fixed O&M component of the capacity payment represents the fixed costs of all the staff for O&M, Power Plant administration, security, transportation, overheads, office costs, professional fees such as audit, tax and legal, as well as some minor fixed operational costs such as environmental monitoring, that do not change with dispatch levels. This also includes the Insurance during operation period

4.7.1.2 ROE and ROEDC

Return on equity (ROE) and return on equity during construction (ROEDC) comprises a return on equity invested giving Return on equity of 22% p.a and an IRR of 20.5% net after deduction of withholding tax. This is based on the fact that as per the Guidelines for the Determination of Tariff for IPPs (November 2005), para 1.4 (c) , it has been mentioned that IRR should be equal yield on 10 years PIB plus a premium of X% to be determined by NEPRA.

For a similar project (Blue Star Energy Limited- 2 MW), the Authority allowed an IRR of 18%. The applicable yield in that case was 10.15% which means, a premium of 7.85% .In case of Lucky Hydropower Project, applicable yield on PIB is 12.7%. With the same premium as was allowed for Blue Star, the IRR requested for approval is 20.5%)

Debt Component

The Debt component is calculated with the following assumptions:

- ✓ Debt is 80% of the Capital Cost of the Project (Including IDC).

- ✓ Repayment in six monthly installments.
- ✓ Grace period equal to the construction period (32 months after Financial Close).
- ✓ Total amount of debt to be arranged from institutions at the bench mark rate of KIBOR (8%) plus a spread of 4.5%.
- ✓ Debt to be repaid in 10 years after Commercial operations date.

In case there is any change in the assumptions, the debt service component and the Tariff shall be adjusted accordingly.

Redemption of Equity

The Project is being developed on BOOT basis. As per the guidelines for the Tariff Determination, IPPs developed on BOOT basis are allowed equity redemption after the completion of debt servicing. The equity is redeemed in equal installments over the remaining term of the PPA. The exchange rate variation shall be allowed in the amount of equity redeemed.

1.ADJUSTMENTS After Finalization of EPC and at COD for EPC Cost .

After the finalization of EPC, Adjustment in EPC price shall be made as per the actual numbers in accordance with the mechanism for Determination of Tariff , mentioned at para 3.1 above. The agreed cost shall be further adjusted for reopeners at the time of COD in accordance with the mechanism mentioned at para 3.1 above.

2. ADJUSTMENTS AT COMMERCIAL OPERATIONS DATE

Total US Dollar Project Cost shall be updated at COD. Debt service, Return on Equity and Return on Equity during Construction shall be adjusted on account of actual variation in debt and equity drawdown, actual interest during construction and financing costs/fees, actual custom duties and taxes and Insurance during Construction. Once adjusted, the Debt service, Return on Equity and Return on Equity during Construction shall be updated accordingly and the relevant Capacity Charges calculated thereon.
The relevant reference tariff components shall be adjusted at COD on account of variation in US$/PKR parity.

3. MODIFICATION / ADDITIONS TO BE TREATED AS PASS THROUGH

The monetary impact of all or any modifications or additions required by the Power Purchaser that are not considered in the Project shall be treated as pass-through.

4. ESCALATIONS AND INDEXATIONS

After the COD, the tariff tables provided will be indexed to factors as described above and the Reference Exchange Rates. The details are provided herein below:

5.1 Inflation Factors

The following components are subject to inflation factors :

Water Use Charges	Pakistan WPI
ROE/ ROEDC/ Withholding tax @7.5%	US CPI

Variable O&M – Local	Pakistan WPI
Variable O&M -Foreign	European CPI/US CPI
Fixed O&M - Local	Pakistan WPI
Fixed O&M - Foreign	European CPI/US CPI

5.2 Currency Indexation

The following components are subject to exchange rate indexation. The Reference Exchange Rates are 84 PKR/1 US$, 1.41 US$/1 € and 114.24 PKR/1 €.

Variable O&M – Foreign	PKR/ Euro and USD exchange rate
Fixed O&M – Foreign	PKR/€ and USD exchange rate
Insurance	PKR/US$ and on the basis of actual insurance payment
ROE /ROEDC	PKR/US$ exchange rate
Withholding tax @7.5%	PKR/US$ exchange rate
Foreign Loan principal	PKR/US$ exchange rate
Foreign Loan interest	PKR/US$ exchange rate

5.3 Interest Rate Indexation
The following components are subject to interest rate indexation:

Financing cost on working capital	3 months KIBOR+ 4.5%
Interest Charge Local Loan	6months KIBOR+ 4.5%
Interest Charge Foreign Loan	6 months LIBOR + 4.5%

6. Assumptions

1. Anticipated average mean site conditions, used in calculation, for the net output and heat rate are:

- Annual Plant Factor -50%
- Installed Capacity -20 MW
- Auxiliary Consumption -0.4 MW
- Contract Capacity -19.6 MW

1. Construction period of 32 months has been assumed as envisaged in the feasibility study..
2. Every maintenance cycle shall be as per manufacturer's recommendations.
3. Output degradation curve from the manufacturer and EPC contractor shall be applied to adjust

the Capacity Cost component of the tariff.

4. Power Purchaser shall be responsible for financing, constructing, operating and maintaining the interconnection to the grid and the electronic communication equipment. However the cost of transmission line has been included in the cost estimates based on information provided by GEPCO. The variation in the estimates considered in the Petition and those which will be actually needed by GEPCO would be accommodated. The arrangement for financing of transmission line would be on a supportive basis as GEPCO has mentioned scarcity of funds. However, the Project Company would be not be liable for mandatory arrangement of funds for the transmission line.

5. A tolerance of +1.5% in dispatch is assumed.

6. Reference exchange rate (PKR/US$) is taken PKR 84/US$ and US$1.41/Euro..

7. The tariff does not incorporate any inflation.

8. No cost of utilizing NTDC telecom media is assumed. Any costs incurred with regards thereto shall be treated as 'pass through'.

9. Main energy meter , CT's and PT's and electronic recorder for continuous recording of readings will be provided by NTDC at its own cost.

10. The Tariff shall be adjusted for the actual insurance costs.

    Payments from Power Purchaser have been assumed as follows:

    - 70% of Capacity Payment will be paid on or before 30th day of the month following the day invoice is received by Power Purchaser

    - 30% of Capacity Payment will be invoiced on first day of next month and will be paid on or before 30th day following the day invoice is received by Power Purchaser

    - 100% of EPP and taxes will be paid on 24th day of subsequent month..

    - All other pass-through items will be paid on or before 30th day following the day invoice is received by Power Purchaser.

11. Project contingency/debt service/O&M/maintenance reserves are not included in tariff calculations. If required by lenders, these shall be adjusted in the tariff. It has been assumed that Lenders will accept letter of credit to satisfy debt service reserve requirement. However, if the lenders would ultimately require cash to fulfill this requirement, the costs and tariff will be adjusted accordingly.

12. 7.5% withholding income tax has been assumed as pass-through (not mentioned in Tariff table) and will be billed on dividend distributions accordingly..

13. No tax on payments to be made to the EPC contractor has been assumed. Any withholding tax, if levied, shall be a "pass through" to the Power Purchaser and will be reimbursed on a monthly basis. Any delay in making payment will carry interest and will be reimbursed by Power Purchaser.

14. Custom duties on imported plant and equipment are assumed to be 5%. Numbers will be trued up to actual at COD. Any change in the custom duties or any other duty or tax on import of equipment and

material will be 'pass through' to the Power Purchaser.

15. Sales Tax on purchase of materials and services after COD by the company has been assumed as fully adjustable against the sales tax on generation and sale of electricity by the Petitioner. If sales tax paid cannot be adjusted, it will be a pass-through to the Power Purchaser under the PPA. Sales Tax or any other taxes paid on services, equipment purchased or imported before COD has also not been assumed.

16. No tax has been assumed on in-house consumption of the electricity.

17. No taxes / levies / charges etc. on construction / maintenance / operation of the plant and use of diesel / gas / electricity at the plant have been factored in the tariff calculation.

18. The tariff table shall be updated at COD in order to correct the tariff according to the prevailing CPI, WPI, KIBOR, LIBOR and exchange rates (PKR/US$ and US$/ € and PKR / €).

19. Actual equity investment profile will be used to up date Return on Equity During Construction, at the time of COD.

Actual IDC, using the actual spread above KIBOR and LIBOR, will be used to update the capital cost at COD Any assumptions on commitment fees, upfront fees, arranger costs and similar charges assumed in the funding plan including PRI, etc will be adjusted at financial close.

20. Funding cost of standby equity and debt has not been assumed Any benefit / concession provided to any other IPP shall also be provided to the petitioner. No cost of LCs has been assumed and any cost thereon shall be Pass through.

21. It is assumed that there will be no income tax, corporate or personal minimum tax or any local/provincial taxes / levies / surcharges etc. Any such taxes will be pass through. No payments on account of Workers Welfare Fund, Workers Profit Participation Fund, presumptive tax/ turnover tax, sales tax, excise duty or other duty, levy, charge, surcharge or other government imposition of a like nature (including without limitation, export tax, octroi) have been assumed and same shall be recovered as a Pass-Through Item whenever and wherever payable by the project within 30 days.

Zakt: For so long as the majority of the shareholders of the Company are non Muslims or non-residents of Pakistan, the Zakat and Ushr Ordinance, 1980, as may be amended from time to time, shall not apply to the Company and the GOP shall exempt the Company from the payment of Zakat. The Zakat and Ushr Ordinance, 1980, as may be amended from time to time, shall not apply to shareholders of the Company who are not Muslims or are not residents of Pakistan and such shareholders shall be exempt from the payment of Zakat on dividends paid by the Company.

22. Any exchange losses on foreign currency loans as a result of devaluation of Reference Exchange Rates shall be capitalized as part of Fixed Assets. In case exchange losses are required to be expensed off in Profit & Loss, any consequential impact on revenues, costs and equity IRR shall be reflected in tariff accordingly.

23. Any change in applicable accounting standards which impact revenues, costs and equity IRR shall be reflected in tariff accordingly.

24. 10% custom duty and central excise has been assumed on purchase of spare parts after COD. Any excess duties and/ or taxes levied on spare parts shall be a "pass through" to the Power Purchaser and will be

reimbursed on a monthly basis.

25. Stamp duty and registration fee @1% has been assumed on principal amount of debt assuming that it will be registered in Islamabad and GOP will grant its consent to a consolidated 1% cost. In case, GOP does not provide its consent for 1% consolidated cost or it is required to be stamped and registered in the province of Punjab then all incremental costs will be a pass-through to the Power Purchaser."

26. During the Term, the Lenders and, as applicable, PRI providers, shall not be subject to taxation (or withholding of tax) in Pakistan on their income from mark-up, interest, and PRI premium in respect of money borrowed or debt incurred by the Company for purposes of the Project or a Restoration of the Plant and from any other financing (including short-term, bridge, working capital and permanent financing facilities) provided to the Company for the Project pursuant to the Financing Documents.

The foregoing tariff petition is most respectfully submitted for and on behalf of the Petitioner:

Signature of the Competent Authority representing Company

Date: October 17, 2011

20. 00 MW Lucky Hydropower Project-Marala Headworks

Executive Summary

FINANCIAL

Exchange Rate USD 1=PKR 84

Tariff Details		
Capacity Payments		
Average (year 1-10) Cents/kW/month	4255.2667	
Average (year 1-10) Rs./kW/month	3574.4240	
Average (Year 11-30) Cents/kW/month	1043.5220	
Average (Year 11-30)Rs./kW/month	876.5585	
Average (Year 1-30) Cents/kW/month	2114.1036	
Average (Year 1-30) Rs./kW/month	1775.8470	
Energy Purchase Price		
Average (year 1-10) Cents/kWhr	0.3718	
Average (year 1-10) Rs./kWhr	0.3123	
Average (year 11-30) Cents/kWhr	0.3718	
Average (year 11-30) Rs./kWhr	0.3123	
Average (year 1-30) Cents/kWhr	0.3718	
Average (year 1-30) Rs./kWhr	0.3123	
Tariff at 50% Plant Factor		
Average year 1-10 (US Cents/KWh)	12.0300	
Average year 1-10 (Rs./KWh)	10.1052	
Average year 1-30 (US Cents/kWh)	6.1638	
Average year 1-30 (Rs./kWh)	5.1776	
Average year 11-30 (US Cents/kWh)	3.2307	
Average year 11-30 (Rs./kWh)	2.7138	
Levelized (1-30) Rs./kWh	7.9912	
Levelized -Cents./kWh	9.5133	
Loan		
Domestic Loan		
Local Portion in the Debt	100.00%	
KIBOR	13.00%	
Spread (as allowed by NEPRA)	4.50%	
Reference Interest rate -KIBOR +4.5%	17.50%	
Repayment Period (Years)	10	
Foreign Loan		
Foreign portion in the debt	0.00%	
LIBOR	6.00%	
Spread (as allowed by NEPRA)	5.00%	
Reference Interest rate	11.00%	
Repayment Period (Years)	10	
Project (MUSD)-Base Cost		
Equity	8.2932	20%
Debt	33.1729	80%
Total Capital	41.4661	100%

Capital Cost (Including Soft Costs)-US $ Million

Equity	8.2932
Debt	33.1729
Total Capital Cost	41.4661
Return on Equity	22.00%

20 MW Lucky Hydropower Project- Marala Headworks



RETURN ON EQUITY & EQUITY REDEMPTION

Equity invested -US $ Million	8.293
Return on Equity	22.0%
Amount of Equity Returned p.a -US $	1,824,507.58
Net Saleable Energy Generation-GWh	85.85
Contract Capacity -Kw	19600.00
Return on Equity-US Cents /kWh without redemption	2.125
Return on Equity- US Cents /KW/month	775.726
Return on Equity-Rs./KW/month	**651.610**

1	651.610	0.893	1.785	0	8.293	2.125	0	0	0
2	651.610	0.893	1.785	0	8.293	2.125	0	0	0
3	651.610	0.893	1.785	0	8.293	2.125	0	0	0
4	651.610	0.893	1.785	0	8.293	2.125	0	0	0
5	651.610	0.893	1.785	0	8.293	2.125	0	0	0
6	651.610	0.893	1.785	0	8.293	2.125	0	0	0
7	651.610	0.893	1.785	0	8.293	2.125	0	0	0
8	651.610	0.893	1.785	0	8.293	2.125	0	0	0
9	651.610	0.893	1.785	0	8.293	2.125	0	0	0
10	651.610	0.893	1.785	0	8.293	2.125	0	0	0
11	619.029	0.848	1.696	0.415	7.879	2.019	148.093	0.203	0.406
12	586.449	0.803	1.607	0.415	7.464	1.913	148.093	0.203	0.406
13	553.868	0.759	1.517	0.415	7.049	1.806	148.093	0.203	0.406
14	521.288	0.714	1.428	0.415	6.635	1.700	148.093	0.203	0.406
15	488.707	0.669	1.339	0.415	6.220	1.594	148.093	0.203	0.406
16	456.127	0.625	1.250	0.415	5.805	1.488	148.093	0.203	0.406
17	423.546	0.580	1.160	0.415	5.391	1.381	148.093	0.203	0.406
18	390.966	0.536	1.071	0.415	4.976	1.275	148.093	0.203	0.406
19	358.385	0.491	0.982	0.415	4.561	1.169	148.093	0.203	0.406
20	325.805	0.446	0.893	0.415	4.147	1.063	148.093	0.203	0.406
21	293.224	0.402	0.803	0.415	3.732	0.956	148.093	0.203	0.406
22	260.644	0.357	0.714	0.415	3.317	0.850	148.093	0.203	0.406
23	228.063	0.312	0.625	0.415	2.903	0.744	148.093	0.203	0.406
24	195.483	0.268	0.536	0.415	2.488	0.638	148.093	0.203	0.406
25	162.902	0.223	0.446	0.415	2.073	0.531	148.093	0.203	0.406
26	130.322	0.179	0.357	0.415	1.659	0.425	148.093	0.203	0.406
27	97.741	0.134	0.268	0.415	1.244	0.319	148.093	0.203	0.406
28	65.161	0.089	0.179	0.415	0.829	0.213	148.093	0.203	0.406
29	32.580	0.045	0.089	0.415	0.415	0.106	148.093	0.203	0.406
30	0.000	0.000	0.000	0.415	0.000	0.000	148.093	0.203	0.406

20 MW Lucky Hydropower Project- Marala Headworks



EQUITY REDEMPTION

Equity Invested -US $ Million	8.293
Redemption Period (Years)	20
Contract Capacity-KW	19600.00
Annual Redemption-US $ Million	0.415
Equity Redemption -US Cents /KW/month	176.301
Equity redemption -Rs /KW/month	**148.093**

Return on Equity During Construction

Accumulated return -US $ Million	3.334
Redemption Period (Years)	30
Contract Capacity -KW	19600.00
Annual payment -US $ Million	0.111
Return on equity -US Cents /KW/month	47.247
Return on equity during construction -Rs. /KW/month	**39.688**
Return on equity during construction -Rs. /KW/hr	**0.054**
Retun on equity during construction at 50%p.f-Rs./kWh	**0.109**

20 MW Lucky Hydropower Project- Marala Headworks

Analysis for the Operation & Maintenanace Charges

O&M -% of Capital Cost p.a	2.0%
Capital Cost-US$ Million	41.47
Amount of O&M p.a- US $	829,321.63
Allocation for Fixed O&M	80%
Amount of Fixed O&M p.a- US $	663,457.30
Allocation for Variable O&M	20%
Amount of Variable O&M p.a- US$	165,864.33
Contract Capacity-Kw	19,600.00

Variable O&M	
Net Saleable Energy Generation-GWh	85.85
Amount of Variable O&M p.a- US $	165,864.33
Variable O&M (Cents /KWh)	0.193

Year	O&M Cents /KWh
1	0.193
2	0.193
3	0.193
4	0.193
5	0.193
6	0.193
7	0.193
8	0.193
9	0.193
10	0.193
11	0.193
12	0.193
13	0.193
14	0.193
15	0.193
16	0.193
17	0.193
18	0.193
19	0.193
20	0.193
21	0.193
22	0.193
23	0.193
24	0.193
25	0.193
26	0.193
27	0.193
28	0.193
29	0.193
30	0.193

Fixed O&M	
Mean Annual Energy Generation-	85.85
Amount of Fixed O&M p.a- US $	663,457.30
Fixed O&M- (Cents/KWh)	0.773

Year	Fixed-O&M Cents /KWh	Fixed O&M. Rs./kW/month
1	0.773	236.949
2	0.773	236.949
3	0.773	236.949
4	0.773	236.949
5	0.773	236.949
6	0.773	236.949
7	0.773	236.949
8	0.773	236.949
9	0.773	236.949
10	0.773	236.949
11	0.773	236.949
12	0.773	236.949
13	0.773	236.949
14	0.773	236.949
15	0.773	236.949
16	0.773	236.949
17	0.773	236.949
18	0.773	236.949
19	0.773	236.949
20	0.773	236.949
21	0.773	236.949
22	0.773	236.949
23	0.773	236.949
24	0.773	236.949
25	0.773	236.949
26	0.773	236.949
27	0.773	236.949
28	0.773	236.949
29	0.773	236.949
30	0.773	236.949

20 MW Lucky Hydropower Project -Marala Headworks



Executive Summary

TECHNICAL

Installed Capacity (MW)	20
Annual Plant Factor	50.00%
Operation Days in a year	365
Mean annual energy generation (GWh)	87.60
Auxiliary losses	2.00%
Auxiliary load-MW	0.40
Contract Capacity-KW	19600.00
Auxiliary losses (GWh)	1.75
Net Saleable Energy (GWh)	85.85
Construction Period (Months)	30
Lead Time before construction start (Years) from feasibility study approval	2
Operation & Maintenance Cost % of Capital Cost	2.00%

20 MW Lucky Hydropower Project- Marala Headworks



Analysis for the Insurance Charges

Insurance Premium -% of EPC	1.35%
EPC Cost-US$ Million	29.52
Amount of insurance p.a- US $	398,478.98
Contract Capacity-Kw	19,600.00
Insurance -Rs./Kw/hr	0.195
Insurance -Rs./Kw/month	142.314
Insurance at 50% p.f -Rs./kWh	0.390

20 MW Lucky Hydropower Project- Marala Headworks

Debt Repayment Schedule

Debt inlcuding IDC- US $ Million	33.173
Repayment Period (years	10
Interest Rate	13.0%
Spread	4.5%
Reference interest rate-KIBOR + 4.5%-P.a	17.50%
Mark Up application	Bi annually
No.of Periods	20
Interest Rate	17.50%
Contract Capacity-Kw	19,600.000
Bi Annual Debt Servicing- USD million	**3.5054**

	1	2.9026	0.602783	3.5054	32.570				3.4300	
1	2	2.8499	0.655527	3.5054	31.915	7.011	2980.790	2503.8636		6.8599
	3	2.7925	0.712885	3.5054	31.202				3.4300	
2	4	2.7301	0.775263	3.5054	30.426	7.011	2980.790	2503.8636		6.8599
	5	2.6623	0.843098	3.5054	29.583				3.4300	
3	6	2.5885	0.916869	3.5054	28.666	7.011	2980.790	2503.8636		6.8599
	7	2.5083	0.997096	3.5054	27.669				3.4300	
4	8	2.4211	1.084341	3.5054	26.585	7.011	2980.790	2503.8636		6.8599
	9	2.3262	1.179221	3.5054	25.406				3.4300	
5	10	2.2230	1.282403	3.5054	24.123	7.011	2980.790	2503.8636		6.8599
	11	2.1108	1.394613	3.5054	22.729				3.4300	
6	12	1.9888	1.516642	3.5054	21.212	7.011	2980.790	2503.8636		6.8599
	13	1.8561	1.649348	3.5054	19.563				3.4300	
7	14	1.7117	1.793666	3.5054	17.769	7.011	2980.790	2503.8636		6.8599
	15	1.5548	1.950612	3.5054	15.818				3.4300	
8	16	1.3841	2.121291	3.5054	13.697	7.011	2980.790	2503.8636		6.8599
	17	1.1985	2.306904	3.5054	11.390				3.4300	
9	18	0.9967	2.508758	3.5054	8.882	7.011	2980.790	2503.8636		6.8599
	19	0.7771	2.728274	3.5054	6.153				3.4300	
10	20	0.5384	2.966998	3.5054	3.186	7.011	2980.790	2503.8636		6.8599

20 MW Lucky Hydropower Project- Marala Headworks

Tariff Table -in Rs.

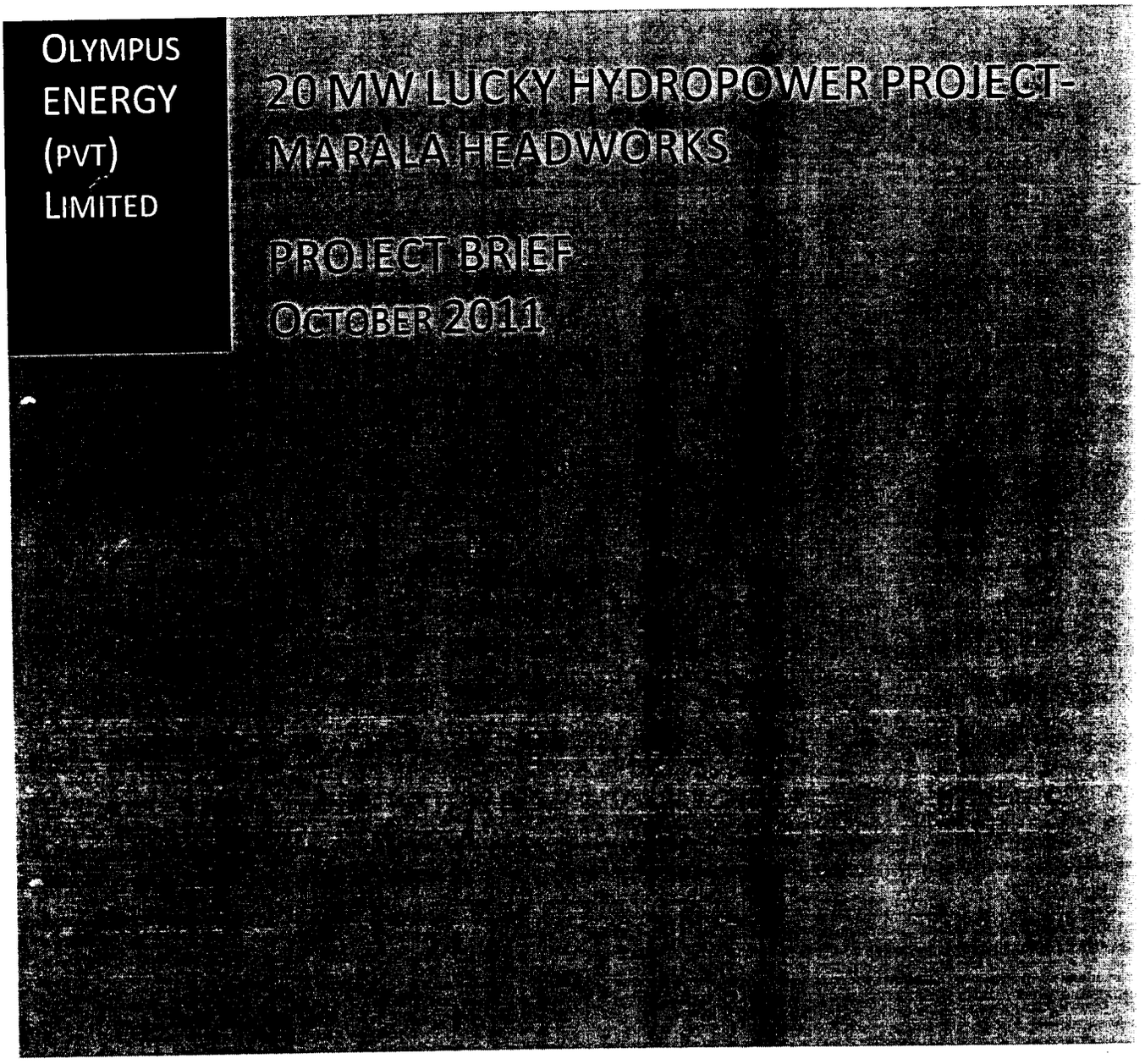
	Capacity Payments (Rs./KW/h)								Tariff at 50% Plant Factor (Rs./kWh)	Energy Purchase Price (Rs./kWh)		
1	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
2	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
3	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
4	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
5	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
6	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
7	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
8	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
9	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
10	236.949	651.610	0.000	2503.864	142.314	39.688	3574.424	9.793	10.105	0.162	0.150	0.312
11	236.949	619.029	148.093	0.000	142.314	39.688	1186.073	3.250	3.562	0.162	0.150	0.312
12	236.949	586.449	148.093	0.000	142.314	39.688	1153.493	3.160	3.473	0.162	0.150	0.312
13	236.949	553.868	148.093	0.000	142.314	39.688	1120.912	3.071	3.383	0.162	0.150	0.312
14	236.949	521.288	148.093	0.000	142.314	39.688	1088.332	2.982	3.294	0.162	0.150	0.312
15	236.949	488.707	148.093	0.000	142.314	39.688	1055.751	2.892	3.205	0.162	0.150	0.312
16	236.949	456.127	148.093	0.000	142.314	39.688	1023.171	2.803	3.116	0.162	0.150	0.312
17	236.949	423.546	148.093	0.000	142.314	39.688	990.590	2.714	3.026	0.162	0.150	0.312
18	236.949	390.966	148.093	0.000	142.314	39.688	958.010	2.625	2.937	0.162	0.150	0.312
19	236.949	358.385	148.093	0.000	142.314	39.688	925.429	2.535	2.848	0.162	0.150	0.312
20	236.949	325.805	148.093	0.000	142.314	39.688	892.849	2.446	2.758	0.162	0.150	0.312
21	236.949	293.224	148.093	0.000	142.314	39.688	860.268	2.357	2.669	0.162	0.150	0.312
22	236.949	260.644	148.093	0.000	142.314	39.688	827.688	2.268	2.580	0.162	0.150	0.312
23	236.949	228.063	148.093	0.000	142.314	39.688	795.107	2.178	2.491	0.162	0.150	0.312
24	236.949	195.483	148.093	0.000	142.314	39.688	762.527	2.089	2.401	0.162	0.150	0.312
25	236.949	162.902	148.093	0.000	142.314	39.688	729.946	2.000	2.312	0.162	0.150	0.312
26	236.949	130.322	148.093	0.000	142.314	39.688	697.366	1.911	2.223	0.162	0.150	0.312
27	236.949	97.741	148.093	0.000	142.314	39.688	664.785	1.821	2.134	0.162	0.150	0.312
28	236.949	65.161	148.093	0.000	142.314	39.688	632.205	1.732	2.044	0.162	0.150	0.312
29	236.949	32.580	148.093	0.000	142.314	39.688	599.624	1.643	1.955	0.162	0.150	0.312
30	236.949	0.000	148.093	0.000	142.314	39.688	567.044	1.554	1.866	0.162	0.150	0.312
NPV								**61.85**	**64.37**			**2.52**
Levelized -Rs/KWh									**7.9912**			**0.3123**
Levelized -Cents /KWh									**9.5133**			

20 MW Lucky Hydropower Project
Summary of the Tariff

Capacity Payments		**Average Year (1-10)**		**Average Year (11-30)**		**Average year (1-30)**	
Sr. No.	Component	Rs./KW/hr	Rs./KW/Month	Rs./KW/hr	Rs./KW/Month	Rs./KW/hr	Rs./KW/Month
1	Fixed O&M	0.3246	236.9490	0.3246	236.9490362	0.3246	236.9490362
2	Return on Equity	0.8926	651.6098495	0.4240	309.5146785	0.5802	423.5464022
3	Debt Servicing	3.4300	2503.8636	0.0000	0.0000	1.1433	834.6212
4	Insurance	0.1950	142.3139	0.1950	142.3139	0.1950	142.3139
5	Return on Equity during construction	0.0544	39.6877	0.0544	39.68767265	0.0544	39.6877
6	Equity redemption	0.0000	0.0000	0.2029	148.0931476	0.1352	98.7288
7	Capacity Payment	4.8965	3574.4240	1.2008	876.5585	2.4327	1775.8470
Energy Payments		**Average Year (1-10)**		**Average Year (11-30)**		**Average year (1-30)**	
Sr.No.	Component	Rs./KWhr		Rs./KW/hr		Rs./KW/hr	
1	Variable O&M	0.1623		0.1623		0.1623	
2	Water Use Charges	0.15		0.15		0.15	
	Energy Payments	0.3123		0.3123		0.3123	
Levelized Tariff							
Rs./kWh (Year 1-30)	**7.9912**						

9.513319186

Annexure-A


OLYMPUS ENERGY (PVT) LIMITED
20 MW LUCKY HYDROPOWER PROJECT-
MARALA HEADWORKS
PROJECT BRIEF
OCTOBER 2011

EXECUTIVE SUMMARY

1. INTRODUCTION

1.1 General

Pakistan is facing acute power shortage due to lack of attention in hydropower sector for last three decades. Presently, the gap between power supply and demand is more than 5000 MW. It is feared that the situation will become even worse in coming years. The electricity demand is increasing @ 8 to 9% per annum. In order to reduce the gap and to exploit all the potential resources for power generation, the Government of Pakistan has decided to involve private sector and set up Private Power Infrastructure Board (PPIB) and NEPRA for the implementation of power generation projects in the private sector and to regulate tariffs and to safeguard the interests of all stake holders. Like wise, the Government of Punjab has set up the Punjab Power Development Board (PPDB) under Irrigation and Power Department (IPD) in 1995 to develop low head hydropower projects at barrages and on canal falls.

Marala Barrage site was identified in 1980 through ranking studies carried out by WAPDA, on the Irrigation network of the country. The PPDB thus considered a Low Head Hydropower Project at Marala Barrage and issued a Letter of Interest (LOI) in favor of M/S Olympus Energy (Pvt.) Ltd. in December, 2007 vide No. MD-PPDB/H-5/489 dated 3-12-2007 to carry out feasibility study by the name of Lucky Hydropower Project at Marala Barrage.

1.2 Description of the Project Area

The Proposed Project area is located in District Gujrat along the right bank of Marala Barrage. The project site is about 48 kilometers upstream of Khanki headwork, 45 km from Gujrat city, 25 kilometers from Sialkot city and about 145 km from Lahore, the capital of Punjab province.

1.3 Marala Barrage

The existing Marala Barrage was constructed in the period 1965-68 under the Indus water treaty of September, 1960. The Barrage length is **1363m** with 66 No. bays including 13 nos. under sluices on the left side and 07 nos. on the right side. The discharge capacity of the Barrage is **31149 m^3/s (1,100,000 cusecs).** Upper Chenab Canal (UCC) having design discharge 467.30 m^3/s (16,500 cusecs) and

Marala Ravi (MR) Link Canal with 623 m^3/s (22,000 cusecs) off-take on its left bank.

2. TOPOGRAPHIC SURVEY

Contract for the survey work was awarded to M/S Geomatics & Engineering Services (Pvt.) Limited in April 2008 for Topographic Survey work over an area of 1243 acres at a scale 1:1000 and contour interval of 0.25 m which was completed in May 2008. Further, to examine the hydraulic conditions at the tail race on its discharging into the river, an additional survey of about 80 acres was carried out in July 2008.

3. HYDROLOGY

3.1 Climate

The climate of the project area is generally hot and dry in summer (April to September) and cold in winter (October to February). Mean annual rainfall for the period of twenty years (1981-2001) is **1045 mm**. Most of rainfall occurs in summer season (80%), and varies from 9 mm in November to 324 mm in August. Average relative humidity varies from 76 to 87%. In summer, temperature in the project area varies between 22 to 44.5°C and in winter, falls even below 0°C. Wind speed varies from 0.2 to 2 knots in November and May respectively.

3.2 Hydrological Data and Processing

Downstream flow data, upstream and downstream water levels on daily basis were collected from Office of Executive Engineer (Marala Barrage Division) I&P, Department, Government of Punjab. Daily flows and head across (Upstream El. – Downstream El.) are computed from Daily data. Daily downstream flow data for period of 1980-2007 (27years) is used for analysis. Mean 10-Daily flow indicates that above 600 cumecs is available for month of May to mid September and 100-250 cumecs for Jan to March , while in October to December downstream releases through Marala Barrage is less than 100 cumecs.

Pond is operated at 247.5 m in summer and 246.3m in winter. Head across the Barrage varies during the year. In the summer, it goes down to 1.5 m in high flood events, whereas in winter it is at max of 5.5 m.

In order to ascertain flow availability through out the year, flow duration curve is drawn on 10-Daily downstream flow for period of 1980-2007. It can be

concluded that 30, 35 and 40 % of time flow availability is 700, 600 and 500 cumecs respectively.

3.3 Barrage Operation

The operation of the barrage is discussed as follows:

3.3 (a) Present Pond Operation

The barrage maximum pond level is regulated at **247.50 m (812 ft).** The 13 nos. under sluices on the left takes care of the siltation in front of the two head regulators for UCC and MR link. There are 07 nos. under sluices on the right for the purpose of the bed load and silt clearance at the barrage. Operation of the under sluices and 46 nos. standard bays of the barrage are regulated in the high / low flows such that the problematic deposition of siltation may not hamper or cause damage to the irrigation system.

The barrage operation, besides maintaining the requisite water supplies in the two canals on the left bank, will also take care of the power house demand (proposed in the system).

The barrage pond level in the proposed operation will be kept at **247.5m (812 ft)** and **246.3m (808ft)** as is required for the water supplies of the two canals in summer and winter season. The gate operation of under sluices on the left and right as well as in the standard bays of the barrage will be regulated in the flood and low flow seasons such that the sediments and the bed load clearance u/s of the Barrage may not remain an issue.

3.4 Power and Energy Potential

Power and energy potential is estimated for 10-daily historic record of downstream flow and upstream/downstream water levels. Estimated power potential, annual energy and plant factor are 20 MW (Contract Capacity 19.6 MW), 87.4 GWh (85.85 GWh after considering auxiliary consumption) and 50 % respectively against design discharge of 580 cumecs for power releases.

4 GEOLOGY, GEOTECHNICS AND SEISMICITY

4.1 Geology

Geologically the project area is part of Upper Indus Basin and tectonically falls in the north Punjab monocline which is a subdivision of Indus Platform and Foredeep zone. The rocks in the region are dominantly sedimentary (sandstone, shale, limestone and silt stone). Besides, metamorphic and igneous rocks are also present in the region.

Geological mapping of the project area including Pond area, headrace channel, powerhouse, tailrace channel, and Borrow areas of construction material were

carried out on scale 1:4000. Rocks are not exposed in the project area, only alluvial deposits such as Clayey Silt, Silty Sand and Sandy Silt with Clay and Gravels are at places. All the alluvial deposits were deposited by the Chenab River and its tributaries in the area and are of recent age.

4.1.1 Geotechnics

4.1.2 Geotechnical Investigation

The geotechnical investigation program for Lucky Hydropower Project Marala Barrage comprises 105 m of exploratory boreholes at Proposed Power House area and Bridge location. Test pits were also excavated to gather information about borrow areas. Using Straight Rotary Drilling Technique and representative sampling, Insitu testing including Permeability and SPTs at different intervals were also carried out.

At the power house location, a thin layer of Silty Clay at the top is observed in BH-05 which is overlaid by poorly graded Sand and the layer comprising same material is observed in the boreholes BH-3 up to 6 m depth. Below 6m, a bed of Silty Sand with Gravel about 4 m thick has been encountered which is overlaid by 2 m thick layer of Silty Clay with Sand. About 12 m from the NSL, 2-3 m thick layer of poorly graded sand overlaid by 10-13 m thick Lean Clay layer is present. Below this, to a drilled depth of 40 m, Silty Sand is observed.

At the bridge location, typical alluvial deposition is observed during drilling and subsequent laboratory testing on selected samples of borehole BH-2.The strata comprises layers of Silty Clay (1.5m), Poorly Graded Sand (11m), Lean Clay (1.5m) and Sandy Silt (1.5m) overlaid by alternate layers of Poorly Graded Sand and Lean Clay to a maximum drilling depth of 35 m.

The ground water level is observed at an average depth of about 0.6 m below the ground surface.

The average SPT Values for the power house corresponding to Layer 1 from depth 0-08 m, Layer-2 from 8-28 m, and Layer-3 from depth 28-40 m are 04, 19 and 48 whereas corrected N70 values are 2, 14 and 30 respectively. For the bridge location, these values corresponding to Layer 1 from depth 0-06 m, Layer-2 from 06-13 m, Layer-3 from 13-18 m, Layer-4 from 18-28 m and layer-5 from 28-40 m are 08, 22, 30, 42 and 62 respectively.

In order to estimate the quantity of dewatering and recharge rate of the power house pit, field permeability testing using constant head technique at the selected

depths were conducted under the supervision of experienced geologist. The permeability values ranging from 7.70 x 10-7 cm/sec to 9.96 x 10-4 cm/sec.

The value of 6.00 x 10-4 cm/sec has been adopted for evaluation of dewatering requirement.

Grain size analysis tests carried out on SPT/UDS samples of the soils collected from boreholes at proposed power house and bridge location, indicate that percentage passing No.200 sieve as 2% to 20% in 16 samples (majorly comprising SP/SM, according to USCS) whereas percentage of more than 65 % for fifteen (15) samples (majorly comprising CL, according to Unified Soil Classification System).

Atterberg limit tests were performed on the fraction of clay passing No. 40 ASTM sieve. The liquid limit was determined using Casagrande apparatus. The tests have indicated that the value of liquid limit ranges from 28% to 42 % with an average value of 34% whereas plasticity index ranges from 8% to 20% with an average value of 14%.

For determining unit weight of the foundation soil, seven (7) tests have been carried out on selected samples. For CL (Lean Clay bed) the value of unit weight ranges from 18.47 KN/m3 to 19.91 KN/m3 with an average value of 19.14 KN/m^3. Whereas a value of 18.87 KN/m3 is reported for SP (poorly graded sand bed).

For determining sp. gravity of foundation soil, eight (08) tests have been carried out on representative samples. For Lean Clay layer (CL), the value ranges from 2.69 to 2.71 with an average value of 2.69. Whereas a value of 2.63 is evaluated for sand and silty sand subsoil.

Direct shear test were carried out on seven (7) soil samples. For Lean Clay layer (CL) the value of angle of internal friction ranges from 22° to 28° with cohesion intercept ranging from 12 KPa to 26 KPa. The average angle of internal friction of 25° with average cohesion intercept of 17 KPa. Whereas 28° with zero cohesion intercept is for poorly graded sand bed (SP).

Oedometer Consolidation test was performed on three (03) samples for the determination of Compression Index (Cc). The value of Compression Index ranges from 0.0534 to 0.0692 with an average value of 0.06.

4.1.3 Power House Foundation

The power house foundation is about 72.41 m long and 47.34 m wide. On the basis of field and laboratory tests evaluation, the proposed type of foundation for the power house is Mat/ Raft Foundation. The allowable bearing capacity of the soil at the proposed depth of 16 m below ground surface level is 50 KPa (0.5 ton/ft^2). The maximum total settlement (immediate + consolidation settlement) of the mat foundation corresponding to a contact pressure of 2 KN/m^2 is calculated to be 9 mm.

4.1.4 Bridge Foundation

The proposed A.R. Bridge foundation at the project site is Pile Foundation. A pile group comprising 5 piles in row, each 1.83 m apart, will be provided to a depth elevation 209.5 m and diameter of individual pile in the group will be 0.91m. Pile cap will be provided with top elevation of 249.78 m and thickness of 1.22 m. The average length of pile will be 40 m.

4.1.5 Dewatering

The water table at the project area fluctuates between 0.4 m to 1.3 m throughout the year. A two stage dewatering system comprising dewatering well is proposed for the dewatering system. The calculation of total water quantities and the cone of depression have been carried out by using coefficient of permeability equal to 6.0 x 10^{-4} cm /s. A total 49 number of pumps with a capacity 0.5 cusec each will be required in the first stage and in the second stage 35 pumps with capacity of 0.5 cusec each are sufficient for this purpose

4.1.6 Construction Materials

Adequate fine material is available in the vicinity of the project site (clayey silt and silty clay) for the construction of embankment of the channel. The nearest barrow area is within 10 km from the Marala Barrage on Sialkot to Gujrat road near the Kurianwala Village.

Adequate fine to medium grained sand mixed with gravel and gritty material is available in the vicinity of the Project area. The nearest borrow area for the sand to construct the structures at project site; is the Chenab River's flood plain.

The rocks and gravels are not available for the aggregates in the vicinity of the Project Site. The aggregates are available from the established source, Margala

hills (Texila) and Kirana hills (Sargodah) which are about 275Km from Marala Barrage.

Material for stone pitching is available from the two established sources, such as Margala hills (Texila) and Sekhanwali which are in the radius of 300 km from the Marala Barrage.

4.1.7 Seismicity

On the basis of earthquake data, GSP has prepared a map showing seismic Hazard zones of Pakistan. According to the seismic zoning map of Pakistan, the Project area falls in zone-3 liable for moderate to severe damage corresponding to Max. Intensity is 7.5 to 9.6 and seismic coefficient of 0.3g. Accordingly seismic factor 0.3g horizontal ground acceleration with maximum credible earthquake (MCE) is suggested for design of Lucky Hydropower Project and its appurtenant structure and the Operation Basis Earthquake (OBE) is proposed to be 0.15g.

5 ENVIRONMENTAL ASPECTS

The environmental considerations have formed an integral part of the evaluation of the layout and design such that all the potential effects of the project have been mitigated. The project is environment friendly. As the Project will not pose any resettlement or relocation issues therefore only Initial Environmental Examination Studies (IEE) were needed on the project. The land proposed for the project belongs to I&P Department, Government of Punjab which will be transferred according to the prevailing rules and regulations.

6 PROJECT LAYOUT

On the basis of survey of Pakistan maps and maps of Marala Barrage division, downloading from google earth and visits to the site helped us in determining the site for the proposed low head hydropower project on the right bank of Marala Barrage. The left bank could not be considered on account of the two canals; UCC main of 467.30 m3/s (16500 cusecs) and Marala Ravi Link canal of 623 m3/s (22000 cusecs) as well as a proposed low head HPP of 11.5 MW on the UCC about 400m below the head regulator of UCC. An area of 1243 acres was marked for a fresh survey along the right bank of Marala Barrage on scale 1:1000 and contour interval 0.25m. Based on the fresh survey maps, 03 nos. of possible layout alternatives were marked. After its comparative study of merits and demerits, alternative 1 was recommended for further study.

In this layout the headrace channel 705 m long from the power house (PH) and off-takes from upstream nose of the right guide bank (RGB) and runs almost parallel to RGB. The powerhouse is placed on the down stream side of the existing Marala-Gujrat road. The alignment of power channel is planned to cross the road perpendicularly. The Powerhouse will be located about 175 m down stream of Marala - Gujrat Road as well as about 250 m from right guide bank of barrage. This is safe distance to avoid the impact on Barrage during excavation / dewatering at the power house Pit. The tailrace channel 366m long will join River Chenab downstream of barrage. The flow directions of river and TRC downstream are coherent. In case of functioning of the breaching section, strengthening of right embankment has also been considered.

7 PROJECT DESIGN

7.1 Civil Engineering Design

The design parameters, assumptions, results of analysis and purpose of all structures required for the power project were established.

The maximum design discharge of the power channel is taken as **580 m^3/s**. The maximum pond level is **247.50 m.a.s.l**. The top level of embankment has been fixed at **250.00 m.a.s.l** which is almost the same level as that of the existing guide bank. The side slopes of the power channel are 1:2. The bed level for the headrace channel at the intake is 242.40 m.a.s.l and FSL at 247.18 m.a.s.l. Velocity in the HRC will be 1.25 m/s and bed width and depth of water will be 88.39 m and 4.88 m respectively.

The upstream portion of power house up to the length of 50 m, 1.83 m thick stone apron is provided filled with gabions as to control the surges due to sudden shut down of power House.

The top width of both the embankment is kept as 10.75 m equivalent to the top width of guide banks of existing Marala Barrage.

The excavated material will be used for the construction of the embankments. The filled area of the embankments should be compacted to 90% of Modified AASHTO.

The tail race is also designed in accordance with the Manning's approach adopting trapezoidal section. The maximum and minimum water levels at the exit of draft tube or start of tailrace channel is 244.08 m and 241.53 m respectively. The side slope of tail race channel is 2:1 with bed width 88 m and depth of flow is 4.61 m and 2.06 m in maximum and minimum conditions of flow.

As far free board of tailrace is concerned, the top of embankment is kept equal to the top elevation of down stream guide bank, as proposed by the barrage authorities against the historic flows.

The inlet bay consists of reinforced concrete cantilever retaining walls acting as approach transition walls.
The out let bay also consists of concrete slab bounded by concrete cantilever retaining walls. Concrete slab starts at down stream of power house at level 234.40m (invert level of draft tube) and sloping up to level 239.47 m (bed level of tail race). It is hinged to the power house foundation with an expansion joint. Sheet piles are proposed underneath the retaining walls to minimize uplift and dewatering efforts.

An access road is provided from barrage to the power house.

The location of power house is 175 m down Stream of AR Bridge and 250 m away from the R.G.B. parallel to the axis of main barrage. The excavation of power house foundation requires deep well dewatering in order to avoid piping. sheet piles are proposed on upstream and down stream side of foundation slab of power house. The power house structure includes inlet and outlet bay machine hall (six unit blocks) and adjacent service bay. The power house yard is located adjoining to the service bay. A roll-up door allow easy access of vehicles and semi-trailers required for delivery of major electrical and mechanical equipment.

The machine hall extents along the entire unit block and service bay. It houses the turbine and generator and other E&M equipment as generator terminals, AC/DC distribution and oil cooling units. Its floor level is at 250 masl with a length of 99.41 m and a width of 14.69 m. The necessary height is defined by the hoisting requirement and is estimated at being 12.25 m.

The service area comprises the control room, offices and storage areas. The service block is used as platform during erection and maintenance of turbines, generators and other E&M equipment.

Trash racks are installed at the intake at an angle of 76° to facilitate mechanical cleaning and to reduce hydraulic losses.

Stop logs are provided at the intake and at the end of the draft tube in order to facilitate erection, repair and maintenance of turbines. Placing and removing of draft tube stop logs is done by using monorail hoist. Intake stop logs are placed and removed by using a racking machine.

7.2 Hydromechanical Equipments

The proposed turbine for Lucky Hydropower Project is double regulated pit type Kaplan unit of 4.31 m runner diameter having 3 adjustable stainless steel blades. The turbine will be capable of operating within a wide range of head and discharge at maximum efficiency. The unit rated discharge is 96.67 m^3/sec with a design net head of 4.7 m and turbine speed of 90.9 r.p.m. Transmission gear has been proposed between turbine and generator to attain the generator speed of 750 r.p.m. The generator voltage will be 6.6 kV.

7.3 Electrical Equipment - Generators, Transformers, Bus-bar & Auxiliary Equipment

The generators comprise horizontal shaft synchronous type, coupled to the transmission gear and then to the Kaplan turbines. The rating of 06 nos. generators will be of 4167 KVA, 0.8 PF, 750 rpm, 6.6 kv, 3 phase, 50 c/s, speed increaser ratio 1:8.25 and with static exciter.

Three step-up transformers 6.6 / 132 kv, 8.5 MVA are proposed and will connect the 6.6 KV Bus-bar to 132 KV Bus-bar. From 132 KV Bus-bar the power will be transmitted to the nearest Head Marala Grid station at a distance of about 3.5 Kms.

8 PROJECT COST ESTIMATE

The total project construction cost of civil works has been estimated on the basis of cost of other similar projects handled by the Consultants and prevaling Market Rate of September, 2008 for Punjab Province. The capital cost is formulated under three distinct heads.

- The civil works cover preliminary works, power channel, powerhouse and switchyard, tail race channel, offices, residential colony and other items of civil works.
- Supply, transportation, erection, testing and commissioning of electromechanical equipment.
- Development cost, corporate overheads, project management cost, Construction insurance and interest during construction, land and other financing charges.

Quantities for Civil Works have been computed on the basis of Layout Plan developed for the scheme and actual topography of the area. The quantities of excavation were obtained from the cross sections drawn at suitable intervals. Quantities for structures have been calculated from the plans and sections of various component structures. The quantities of electrical and mechanical

equipment have been computed on the selected size and type of generating units and preliminary design.

Rates for different items of Civil Works are estimated from direct cost market rate corresponding to November 2008. The rate of each work has been prepared keeping in view the local conditions and relevant specifications. Rates of E&M works such as turbines, generators, transformers and other related equipment have been taken keeping in view the rates at the on-going similar projects, engineering judgment and consultation with manufacturers & suppliers .

9 CONSTRUCTION PLANNING

The project construction is estimated to be completed in 32 months. The construction will be carried out in three stages: Stage-1 of 10 months, Stage-2 of 14 months and Stage-3 of 8 months. The construction planning of the project is based on the following considerations:

I. No interference with irrigation supply
II. No adverse effect to the existing Barrage Structures
III. Regular supply to the off taking canals during and after construction of the project.

The excavation of the powerhouse and of the headrace and tailrace channels will commence simultaneously. Therefore, installation of equipment required for the erection of dewatering system, diaphragm wall and execution of earthwork has to be undertaken immediately. The reinforcement bending yard and the carpenter's yard including the storage yards will be erected during the final stage of excavation of the powerhouse pit. The erection of these yards has to be completed simultaneously with the completion of the powerhouse pit for starting the concrete works of the powerhouse.

The power house is estimated to have an installed capacity of **20 MW (Contract Capacity 19.6 MW)** and an annual energy output of **85.85 GWh** with available head ranging between 2.3 to 5.3 m. The power house including service bay has a length of 72.41 m and width of 47.34 m. The total length including upstream and downstream concrete floors is 111.28 m.

10 POWER MARKET

Low head hydropower potential sites having no storage capacity are available in the irrigation system of the country. The generating capacity of these projects is low and can not be used to meet the peak demand. The power generated by these units is fed directly to secondary transmission or distribution network. The proposed

Lucky Hydropower Project at Marala Barrage will have a total generation capability of 20 MW (Contract Capacity 19.6 MW) with estimated energy generation of 85.85 GWh per annum. The power station will be connected to the nearest grid station at Marala through a 3.5 km long 132 kV transmission line. Power Purchase Agreement (PPA) will be signed with CPPA for sale of electricity to GEPCO grid.. The energy meter will be located at the power house switchyard for measurement of energy delivered to the system.

10.1 Load Flow Studies

The energy generated by the Lucky Hydropower Project is to be fed to the existing network for its consumption. The generating capacity of the project is of such a level that it can be connected directly to the secondary transmission or distribution system. For resolution of the issue of Power Purchase Agreement and Load Flow Studies, General Manager (CPPA) WAPDA was approached during September, 2008 who further referred the case to Chief Executive Officer GEPCO for consideration of interconnectivity arrangements and its estimated cost.

ECONOMIC AND FINANCIAL ANALYSIS

All costs and benefits attributed to the project are expressed in constant as per June 2009 price level. The discount rate of 10% has been used in this study. Project base cost is estimated to be US$ 34.5555 million. The project life is assumed to be 30 years. The construction period is proposed to be 32 months with annual capital outlay 30%, 45% and 25% during 1st, 2nd and 3rd year respectively. The annual OM&R costs worked out to be US$ 1.257 million per year which is 3% of the financial cost.

Project financial cost is estimated to be US$ **41.4661** million. The OM&R costs worked out as US$ **0.8293** million per annum is about 2 % of the financial cost. For the purpose of the calculation of Levelized tariff, a discount factor of 12% has been used.

SALIENT FEATURES

1)	**Location**	Marala Barrage, Punjab
2)	**River System**	Chenab River
3)	**Main Design Features:**	
	Rated Discharge	580 Cumecs
	Gross Head	5.3 m
	Net Head at rated discharge	4.7 m
	Capacity	20 MW
	Auxiliary load 2%	0.4 MW
	Contract Capacity	19.6 MW
	Annual Energy	85.85 GWh
	Plant Factor	50 %
4)	**Hydro Mechanical Equipment:** **Kaplan Horizontal Pit Type Turbines**	
	No. of Units	6 Nos.
	Rated Discharge per Units	96.67 Cumecs
	Capacity per Unit	3.34 MW
	Runner Diameter	4.316 m
	Approx. Runner Weight	21.7 Tonnes
	Unit Speed	90.9 rpm
	System Frequency	50 Hz
5)	**Power House:**	
	Power House Overall Dimensions:	
	Length	72.41 m
	Width	47.34 m
	Height	29.69 m
	Machine Hall Level	250 masl
	Machine Hall Length	91.41 m
	Machine Hall Width	14.69 m
	Machine Hall Height	12.25 m

6) Trash Racks:

Width	9.06 m
Height	16.66 m
Inclination	76 Degree

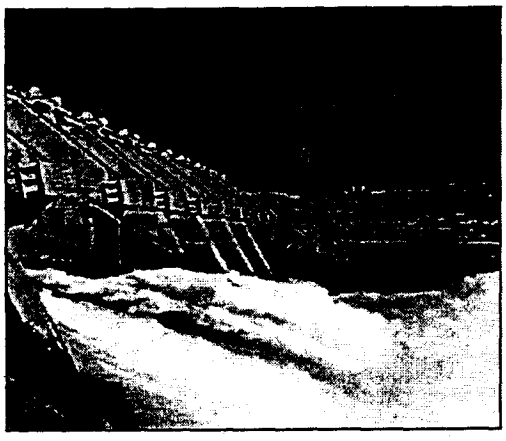
7) Head Race Canal:

Length of Headrace Channel	705 m
Design Discharge	580 m^3/sec
Bed Width (B)	88.39 m
Depth of water (D)	4.88 m
Flow Area	478.38 m^2
Wetted Perimeter	110.19 m
Hydraulic Radius	4.34 m
Critical Depth	1.65 m
Critical Slope	0.00763
Flow Type	Sub-Critical
Profile Type	Mild M-1
Froude No.	0.036
Velocity	1.25 m/sec
Manning's n	0.03
Side Slopes of Headrace Channel (H:V)	2:1 m
Water Surface Slope of Headrace Channel	0.000200 m
Bed Level of Headrace at start of Intake Bay	242.30 masl
FSL of Headrace at P.H.	247.18 masl

8) Tail Race Canal:

Length of Tailrace Channel	366 m
Design Discharge	580 m^3/sec
Bed Width (B)	89 m
Depth of water (D)	4.61 m
Flow Area	454 m^2
Wetted Perimeter	109.65m
Hydraulic Radius	4.13 m
Critical Depth	1.62 m
Critical Slope	0.00765
Flow Type	Sub-Critical
Profile Type	Mild M-2
Froude No.	0.042
Velocity	1.32 m/sec
Manning's n	0.03
Side Slopes of Tailrace Channel (H:V)	2:1 m

Water Surface Slope of Tailrace Channel	0.000237 m
Bed Level of Tailrace at start of Intake Bay	239.47 masl
FSL of Tailrace at Exit.	244.08 masl



20 MW Lucky HYDROPOWER PROJECT

Cost Estimates

Annexure B



SUMMARY OF COST

20 MW Lucky Hydropower Project-Marala Headworks

Sr.No	Description	Local Components		Foreign Components		Total Amount	
		in Million					
		RS	US$	RS	US$	RS	US$
1	**Pre-Construction Stage**						
1.1	Pre-qualifaction	3.000	0.036			3.000	0.036
1.2	Feasibility Study	10.000	0.119			10.000	0.119
1.3	Field Investigations	2.500	0.030			2.500	0.030
1.4	Employer's Office	14.100	0.168			14.100	0.168
2	**Construction Stage**						
2.1	EPC Cost as per feasibility study (Rs 84= USD 1)	**1438.998**	**17.131**	**0.000**	**13.912**	**2607.610**	**31.0430**
2.2	Consultant @ 6% of 2.1	86.340	1.028	0.000	0.000	86.340	1.028
2.3	Employer's Office	75.200	0.895	0.000	0.000	75.200	0.895
3	**Lender Cost**						
3.1	Due Diligence	30.000	0.357		0.250	51.000	0.607
3.2	Registration of Document	5.000	0.060			5.000	0.060
3.3	Advisors	25.000	0.298			25.000	0.298
Project Base Cost as per current exchange rate		**1690.138**	**20.121**	**0.000**	**14.162**	**2879.750**	**34.283**
4. Project Soft Costs							
4.1	Interest During Construction	474.709	5.651	0.000	0.000	474.709	5.651
4.2	Customs Duty -5% of foreign component	58.431	0.696	0.000	0.000	58.431	0.696
4.3	Port Charges and inland transportation-3% of foreign component	35.058	0.417	0.000	0.000	35.058	0.417
4.4	Insurance During Consruction-1.35% of the EPC	35.203	0.419	0.000	0.000	35.203	0.419
4.5	**Project Capital Cost**	**2293.539**	**27.304**	**0.000**	**14.162**	**3483.151**	**41.466**

20 MW Lucky Hydropower Project -Marala Headworks



Return on Equity During Construction & Interest During

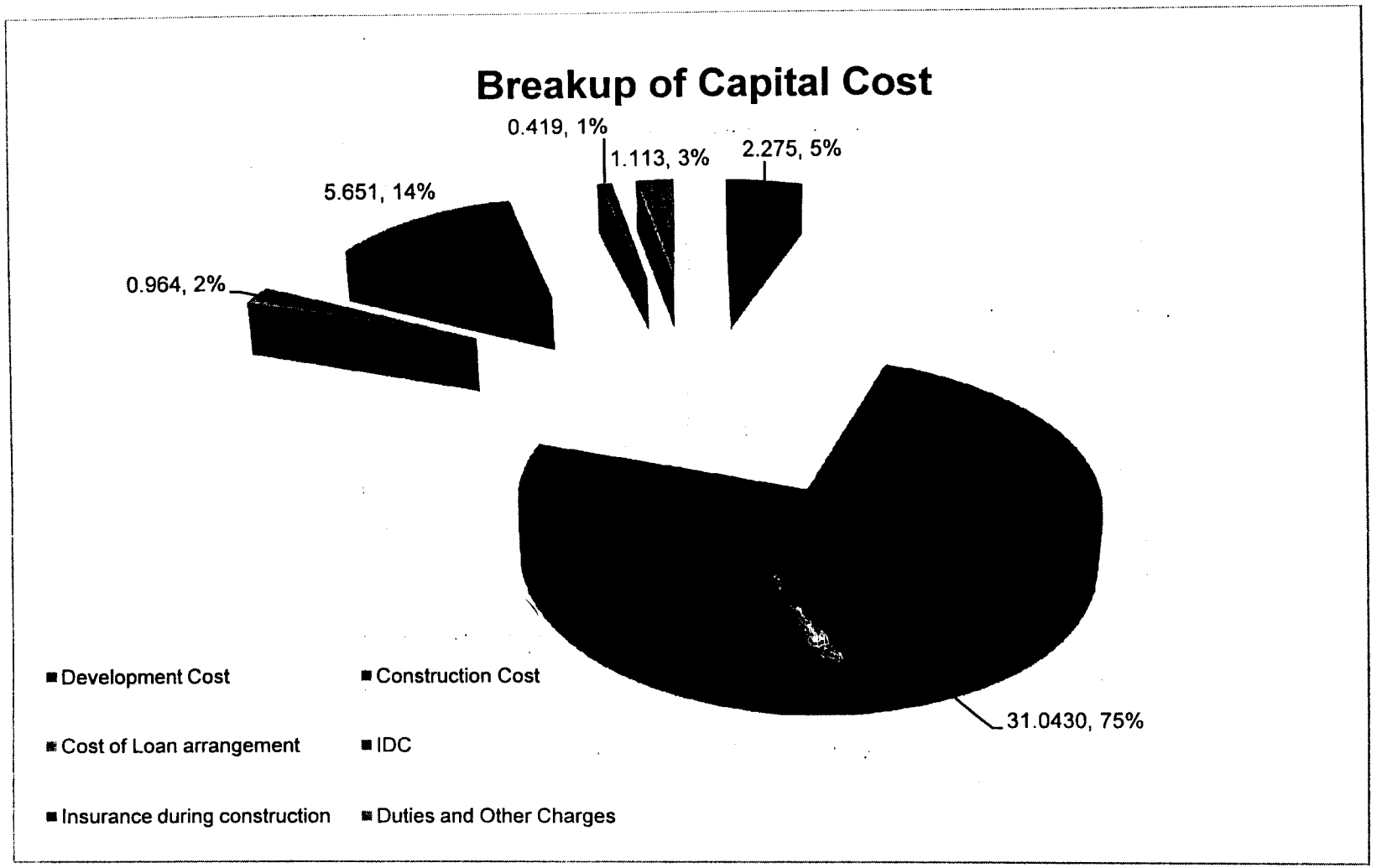
Returns on Equity Invested before Financial Close

Serial No.	Details	Amount US $ Million	Period (Year			
1	Equity invested before Financial Close	1.316666667	Ist	2nd	3rd	4th
		Amount (US $ Million)				
2	Before Financial Close @ 22% p.a		0.29	0.35	0.431	0.526
3	Investment before Financial Close)-A	2.917				

Construction Phase

Project Base Cost	34.283
Equity	20%
Return on Equity invested Before Financial Close	22%
Equity @ Base Cost-B	6.857
Equity to be invested after Financial Close	5.540
Debt	80%
Debt @Base Cost-C	27.426
Interest Rate p.a	0.130
Spread	0.045
Interest Rate p.a	17.500%

	Period					
	0-6 months	**7-12 months**	**13-18 months**	**19-24 months**	**25-30 months**	**Cummulative US $ Million**
Debt						
Debt Phasing	10%	20%	20%	25%	25%	
Disbursement	2.743	5.485	5.485	6.857	6.857	
Interest Rate p.a	17.5%					
Interest During Construction@	0.120	0.490	1.013	1.642	2.386	**5.651**
Accumulated Debt	2.863	8.8383	15.3369	23.8354	33.0775	**33.0775**
Equity During Construction						
Equity Contribution	10%	20%	20%	25%	25%	
Return on Equity P.A	22%					
Amount contributed (US $ Million)	0.554	1.108	1.108	1.385	1.385	
Amount of Return on Equity	0.061	0.190	0.332	0.521	0.731	**1.835**
Net Equity Contributed during Construction	0.615	1.912	3.353	5.259	**7.375**	
Equity Contribution Before Construction plus during construction-US $ Million						**10.291**
Return on EquityDuring Project						**3.435**


Breakup of Capital Cost
0.419, 1%
1.113, 3%
2.275, 5%
5.651, 14%
0.964, 2%
31.0430, 75%
Development Cost
Construction Cost
Cost of Loan arrangement
IDC
Insurance during construction
Duties and Other Charges

20 MW Lucky Hydropower Project- Marala Headworks



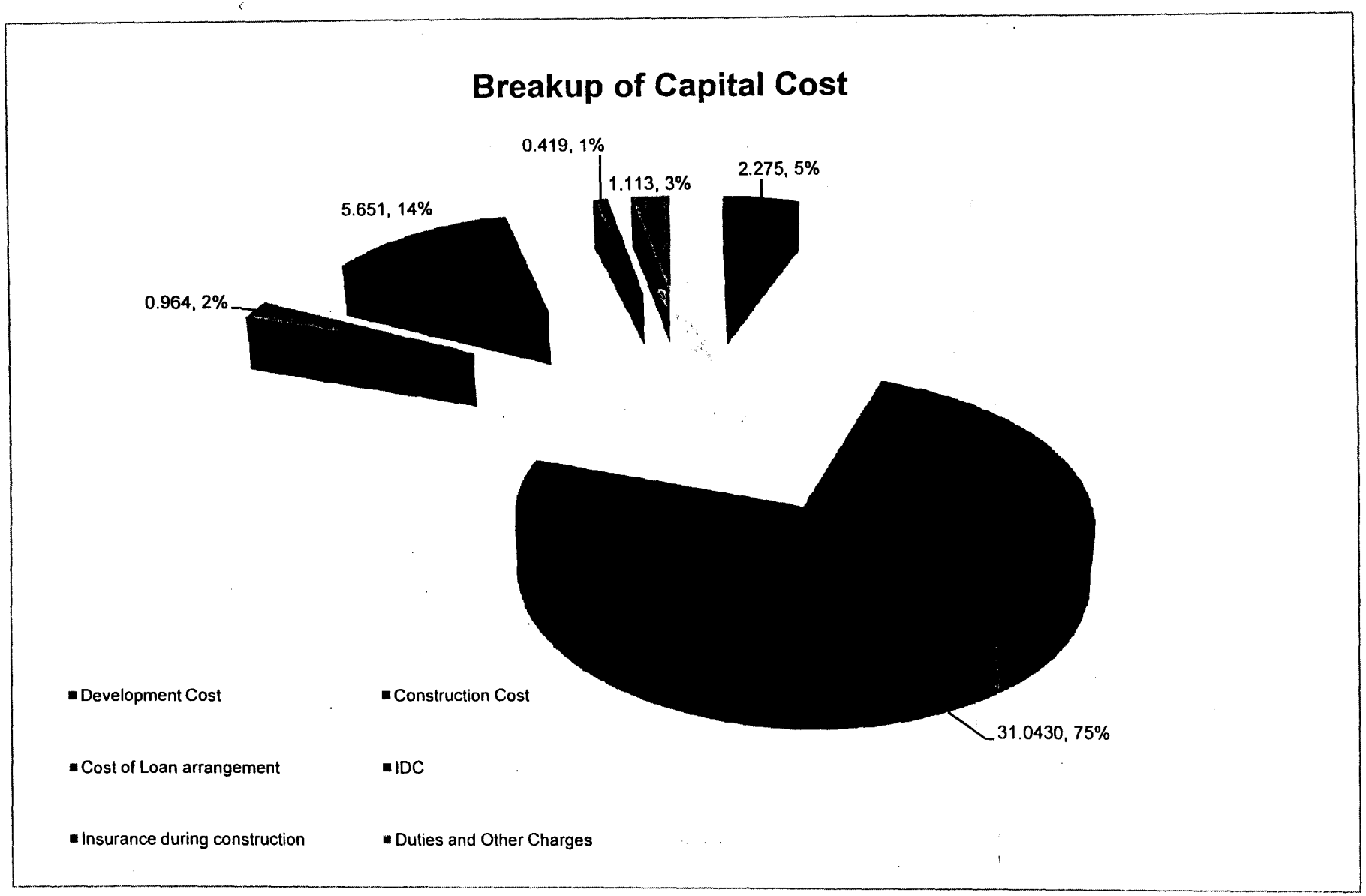
PROJECT CAPITAL COST

Serial No.	Cost Head	Amount (US$ Million)
1	Development Cost	2.275
2	Construction Cost	31.0430
3	Cost of Loan arrangement	0.964
4	IDC	5.651
5	Insurance during construction	0.419
6	Duties and Other Charges	1.113
7	**Capital Cost**	**41.466**

SUMMARY OF EPC COST

20 MW Lucky Hydropower Project-Marala Headworks

Sr.No	Description	Local Components		Foreign Components		Total Amount	
		RS (Million)	US$ (Million)	RS(Million)	US$ (Million)	RS (Million)	US$ (Million)
2.1	Land Acquisition & Preliminary Works including environment mitigation	168.23	2.003	0	0	168.2322	2.0028
2.2	Civil Works	1147.7805	13.6641	0.000	0.000	1147.7805	13.6641
2.3	Mechanicial Works	122.9856	1.4641	0.0000	10.8145	908.4205	12.2786
2.4	Electrical Works	0.0000	0.0000	0.0000	1.1340	95.2560	1.1340
2.5	Transmission Line	0.0000	0.0000	0.0000	1.9635	164.9355	1.9635
Total EPC Cost		**1438.9983**	**17.1309**	**0.0000**	**13.9120**	**2484.6246**	**31.0430**


Breakup of Capital Cost
0.419, 1%
1.113, 3%
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Development Cost
Construction Cost
Cost of Loan arrangement
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Insurance during construction
Duties and Other Charges